OPERATING SYSTEMS PRACTICAL

Experiment: Non-Preemptive Shortest Job First (SJF) CPU Scheduling

1. Aim

To write and execute a C program to implement Non-Preemptive Shortest Job First (SJF) CPU Scheduling, calculate Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT), and display the Gantt Chart.

2. Theory

Shortest Job First (SJF) is a CPU scheduling algorithm in which the ready process having the smallest CPU Burst Time is selected for execution. In Non-Preemptive SJF, once a process gets the CPU, it continues until its CPU burst is completed. The scheduler then selects the shortest job from the processes that have arrived.

SJF can reduce the average waiting time when burst times are known. However, long processes may wait for a long time if shorter processes keep arriving; this is called starvation. This program considers Arrival Time and supports up to 10 processes.

3. Algorithm

1. Start the program.
2. Read the number of processes. Enter Arrival Time (AT) and Burst Time (BT) for each process.
3. Initialize the current CPU time and completed-process count to zero.
4. While all processes are not completed, find an arrived and unfinished process with the smallest Burst Time.
5. If no process has arrived yet, advance the CPU time until a process becomes available.
6. Execute the selected process completely and update the current CPU time.
7. Calculate CT = finishing time, TAT = CT − AT, and WT = TAT − BT.
8. Mark the selected process as completed and store its execution order for the Gantt Chart.
9. Repeat until every process is completed.
10. Display the process table, average TAT, average WT, and Gantt Chart.
11. Stop.

4. C Program

```
#include<stdio.h>

int main()
{
    int n, i, time = 0, count = 0;
    int at[10], bt[10], ct[10], tat[10], wt[10];
    int done[10] = {0};
    int order[10], start[10];
    int shortest;
    float avgwt = 0, avgtat = 0;

    printf("Enter number of processes (1 to 10): ");
    scanf("%d", &n);

    if (n < 1 || n > 10)
    {
        printf("Invalid number of processes. Enter 1 to 10.");
```

```
    return 0;
}

for (i = 0; i < n; i++)
{
    printf("Enter AT and BT of P%d: ", i + 1);
    scanf("%d%d", &at[i], &bt[i]);
}

while (count < n)
{
    shortest = -1;

    for (i = 0; i < n; i++)
    {
        if (done[i] == 0 && at[i] <= time)
        {
            if (shortest == -1 || bt[i] < bt[shortest])
                shortest = i;
        }
    }

    if (shortest == -1)
    {
        time++;
        continue;
    }

    start[count] = time;
    time = time + bt[shortest];

    ct[shortest] = time;
    tat[shortest] = ct[shortest] - at[shortest];
    wt[shortest] = tat[shortest] - bt[shortest];

    done[shortest] = 1;
    order[count] = shortest;
    count++;
}

printf("\\nP\\tAT\\tBT\\tCT\\tTAT\\tWT\\n");

for (i = 0; i < n; i++)
{
    printf("P%d\\t%d\\t%d\\t%d\\t%d\\t%d\\n",
           i + 1, at[i], bt[i], ct[i], tat[i], wt[i]);

    avgwt += wt[i];
    avgtat += tat[i];
}

printf("\\nAverage Waiting Time = %.2f", avgwt / n);
```

```
  printf("\\nAverage Turnaround Time = %.2f", avgtat/n);

  printf("\\n\\nGantt Chart:\\n");
  printf("%d", start[0]);

  for (i = 0; i < n; i++)
  {
     printf("--P%d--%d", order[i] + 1, ct[order[i]]);
  }

  printf("\\n");
  return 0;
}
```

5. Formulas

Term	Formula / Meaning
Completion Time (CT)	The time at which a process finishes execution.
Turnaround Time (TAT)	TAT = CT − AT
Waiting Time (WT)	WT = TAT − BT = CT − AT − BT
Average Turnaround Time	Average TAT = Sum of all TAT values / Number of processes
Average Waiting Time	Average WT = Sum of all WT values / Number of processes

6. Sample Input and Output

Sample input (all processes arrive at time 0):

Number of processes: 4
P1: AT = 0, BT = 6
P2: AT = 0, BT = 8
P3: AT = 0, BT = 7
P4: AT = 0, BT = 3

Expected output:

P	AT	BT	CT	TAT	WT
P1	0	6	9	9	3
P2	0	8	24	24	16
P3	0	7	16	16	9
P4	0	3	3	3	0

Average Waiting Time = 7.00
Average Turnaround Time = 13.00

Gantt Chart:
0 -- P4 -- 3 -- P1 -- 9 -- P3 -- 16 -- P2 -- 24

7. Conclusion

Thus, the C program for Non-Preemptive SJF CPU Scheduling was implemented and executed successfully. The Completion Time, Turnaround Time, Waiting Time, average times, and Gantt Chart were obtained.

Important Note

This program is Non-Preemptive SJF. It selects the shortest Burst Time among processes that have already arrived. If no process is available, CPU time advances until the next process arrives. For the sample input, all Arrival Times are zero, so the execution order is P4, P1, P3, P2.